Severity & Case Classification

Task 4: Classifying adverse drug reactions by seriousness and expectedness

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Objective

To evaluate multiple adverse drug reaction (ADR) cases, classify each by **seriousness** and **expectedness**, note its **severity**, and decide the reporting priority using pharmacovigilance guidelines (ICH E2A).

1. Seriousness

Seriousness is a **regulatory** classification based on the outcome of the event. A case is **serious** if it meets at least one of these criteria:

Seriousness criterion	Example
Results in death	Fatal anaphylaxis
Life-threatening (real risk of death at the time of the event)	Severe lactic acidosis needing ICU care
Requires hospitalisation or prolongs an existing hospital stay	GI bleeding needing admission
Persistent or significant disability or incapacity	Permanent hearing loss
Congenital anomaly / birth defect	Limb defect after exposure in pregnancy
Other medically important event	Seizure treated in casualty without admission

Any case meeting none of these is **non-serious**.

2. Severity is not the same as seriousness

Severity describes the **intensity** of the event (mild, moderate or severe). **Seriousness** describes its **outcome**. A severe headache can be non-serious, while a mild reaction that leads to hospital admission is serious. Reporting timelines depend on seriousness, not severity.

Severity	Meaning
Mild	Noticeable but does not interfere with daily activities
Moderate	Interferes with daily activities; may need treatment
Severe	Prevents daily activities; needs significant treatment

3. Expectedness

A reaction is **expected** if its nature and severity are consistent with the reference safety information: the **product label / package insert / SmPC** for a marketed drug, or the **Investigator's Brochure (IB)** for a drug in a clinical trial. It is **unexpected** if it is not listed, or if it is listed but this case is more severe or specific than described.

4. Classification matrix and action

	Expected	Unexpected
Serious	Report promptly; monitor frequency for a change in pattern	Highest priority. In a clinical trial this is a **SUSAR**, reported to the regulator within 15 calendar days (7 days if fatal or life-threatening)
Non-serious	Lowest priority; routine reporting and periodic review	Routine reporting; review as a possible new signal and label update

Case analysis

Case 1: Amoxicillin: skin rash and itching

Patient	Female, 45 years
Drug	Amoxicillin 500 mg three times daily (marketed)
Event	Itchy red rash on arms and trunk on day 3; settled with an antihistamine after stopping the drug. Treated as an outpatient.

Assessment	Finding
Seriousness	None of the criteria met. No hospitalisation, no threat to life.
Severity	Mild
Expectedness	Skin rash and pruritus are listed in the amoxicillin label.

Classification: Non-serious, Expected
Action: Routine report to PvPI; record penicillin allergy in the patient's notes.

Case 2: Ibuprofen: gastrointestinal bleeding needing hospitalisation

Patient	Male, 64 years
Drug	Ibuprofen 400 mg three times daily for knee pain (marketed)
Event	Black tarry stools and fall in haemoglobin after 10 days; admitted, transfused and treated with a proton pump inhibitor.

Assessment	Finding
Seriousness	Serious: required hospitalisation.
Severity	Severe
Expectedness	GI bleeding is listed in the ibuprofen label as a known NSAID risk.

Classification: Serious, Expected
Action: Report promptly to PvPI as a serious ADR; stop NSAID and use a safer analgesic.

Case 3: Metformin: lactic acidosis

Patient	Female, 71 years, type 2 diabetes
Drug	Metformin 1 g twice daily (marketed)
Event	After an episode of dehydration with worsening kidney function, she developed vomiting, rapid breathing and confusion. Blood lactate markedly raised; admitted to ICU.

Assessment	Finding
Seriousness	Serious: life-threatening and required hospitalisation.
Severity	Severe
Expectedness	Lactic acidosis is listed in the metformin label as a rare but serious warning, especially with kidney impairment.

Classification: Serious, Expected
Action: Report promptly to PvPI; stop metformin; review kidney function before any future prescribing.

Case 4: Atorvastatin: raised liver enzymes

Patient	Male, 52 years
Drug	Atorvastatin 20 mg once daily (marketed)
Event	Routine blood test at 8 weeks showed ALT about twice the upper limit of normal. No symptoms; normal bilirubin.

Assessment	Finding
Seriousness	None of the criteria met. Found on routine testing, no symptoms.
Severity	Mild
Expectedness	Raised liver enzymes are listed in the atorvastatin label.

Classification: Non-serious, Expected
Action: Routine report; repeat liver tests and review the dose.

Case 5: Investigational drug XR-101: metallic taste

Patient	Female, 40 years, clinical trial participant
Drug	XR-101 (hypothetical investigational drug, Phase II trial)
Event	Persistent metallic taste from week 2; no effect on eating or daily activities.

Assessment	Finding
Seriousness	None of the criteria met.
Severity	Mild
Expectedness	Taste disturbance is **not listed** in the Investigator's Brochure.

Classification: Non-serious, Unexpected
Action: Record as an adverse event in the trial database; include in periodic safety review as a possible new finding.

Case 6: Investigational drug XR-101: seizure needing hospitalisation

Patient	Male, 33 years, clinical trial participant, no history of epilepsy
Drug	XR-101 (hypothetical investigational drug, Phase II trial)
Event	Generalised tonic-clonic seizure on day 18; admitted for observation and investigations. Brain imaging and blood tests normal.

Assessment	Finding
Seriousness	Serious: required hospitalisation (and a medically important event).
Severity	Severe
Expectedness	Seizure is **not listed** in the Investigator's Brochure.

Classification: Serious, Unexpected: a **SUSAR** if a causal link is suspected
Action: Highest priority. Expedited report to the regulator (CDSCO) and ethics committee within 15 calendar days; inform investigators; review whether the trial needs changes.

Summary of classification

#	Drug	Adverse reaction	Seriousness	Severity	Expectedness	Priority
1	Amoxicillin	Skin rash, itching	Non-serious	Mild	Expected	Routine
2	Ibuprofen	GI bleeding (hospitalised)	Serious	Severe	Expected	High
3	Metformin	Lactic acidosis	Serious	Severe	Expected	High
4	Atorvastatin	Raised liver enzymes	Non-serious	Mild	Expected	Routine
5	XR-101	Metallic taste	Non-serious	Mild	Unexpected	Routine; signal review
6	XR-101	Seizure (hospitalised)	Serious	Severe	Unexpected	Highest (SUSAR)

Clinical impact

Of the six cases, three are serious and three are non-serious; four are expected and two are unexpected. The two serious expected cases (ibuprofen and metformin) are well-known risks, so the focus is on prompt reporting and prevention: avoiding NSAIDs in high-risk patients, and checking kidney function before metformin. The serious unexpected seizure with XR-101 has the greatest impact, because it may reveal a new safety risk of the drug and requires expedited reporting.

Key learning points

- Seriousness depends on the outcome (death, life-threatening, hospitalisation, disability, congenital anomaly, medically important), not on how intense the reaction felt.
- Severity (mild, moderate, severe) describes intensity and does not decide reporting timelines.
- Expectedness is always judged against the reference document: the product label for marketed drugs, the Investigator's Brochure for trial drugs.
- Serious and unexpected cases carry the highest priority and the shortest reporting timelines.

Note: the cases are hypothetical and written for learning purposes. XR-101 is a fictional drug.